

E-Commerce Sales & Profit Analytics Dashboard

An End-to-End Business Intelligence Project using Excel, SQL Server, Power BI Desktop & Power BI Service

Power BI Dashboard	SQL Server Database	Excel Data Source	DAX Advanced	Status Completed
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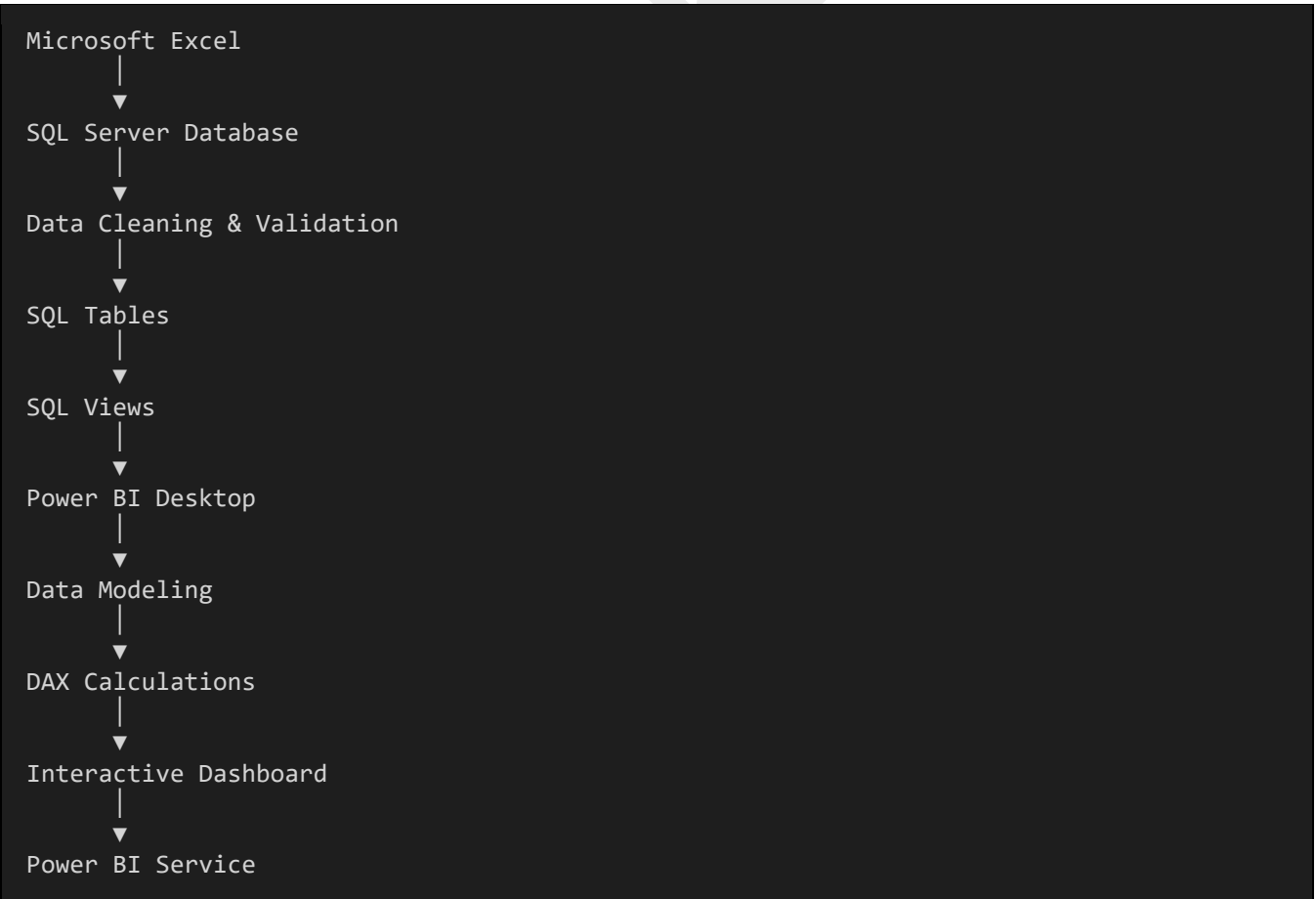
Project Overview

This project demonstrates an end-to-end Business Intelligence solution by transforming raw Excel data into an interactive Power BI dashboard through SQL Server.

The project follows an enterprise BI workflow where Excel data is loaded into SQL Server, transformed using SQL, modelled in Power BI, and finally published to Power BI Service for interactive reporting.

The dashboard provides actionable business insights into sales performance, customer behavior, product profitability, and time-based trends while implementing advanced Power BI development techniques.

Solution Architecture



Technology Stack

- Microsoft Excel
- SQL Server
- SQL Server Management Studio (SSMS)
- SQL Views
- Power BI Desktop
- Power BI Service
- Power Query
- DAX
- Data Modeling

Dataset

The project uses three datasets:

Dataset	Description
Orders	Sales Transactions
Returns	Returned Orders
People	Regional Managers

SQL Server Implementation

Database Creation

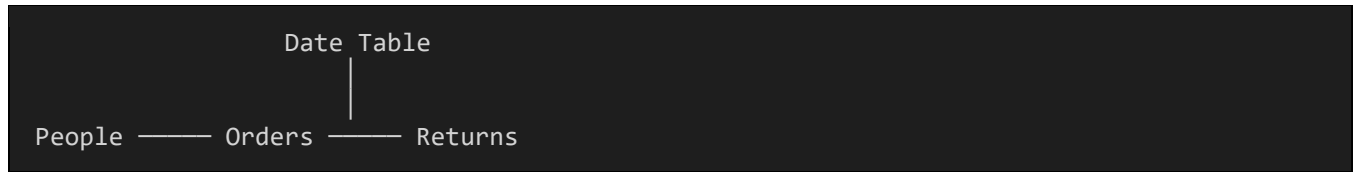
- Created SQL Database
- Created Fact & Dimension Tables
- Imported Excel Worksheets
- Performed Data Validation
- Created SQL Views
- Optimized Queries for Reporting

SQL Concepts Used

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- INNER JOIN
- LEFT JOIN
- Aggregate Functions
- CASE Statement
- Views
- Data Validation Queries

Power BI Data Model

A Star Schema data model was implemented.



Relationships

- Orders → Date Table
- Orders → Returns
- Orders → People

Date Table

A dedicated Date Table was created to support Time Intelligence calculations.

Includes

- Date
- Year
- Quarter
- Month
- Month Number
- Day
- Weekday
- Weekend Flag

Time Intelligence functions were built using this Date Table.

DAX Measures

Developed advanced DAX measures including:

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Total Quantity
- Average Order Value
- Average Discount
- Average Shipping Cost
- Profit Margin %
- Return Rate %

- Orders YTD
- Sales YTD
- Profit YTD
- Previous Year Sales
- Rolling 12 Months
- Dynamic Top N Products
- Dynamic Bottom N Products
- Dynamic KPI Measures

Advanced Power BI Features

- ✓ Data Modeling (Star Schema)
- ✓ Power Query Transformations
- ✓ Date Table
- ✓ Time Intelligence Functions
- ✓ Dynamic DAX Measures
- ✓ Bookmarks
- ✓ Page Navigation
- ✓ Drill Down
- ✓ Drill Through
- ✓ Report Tooltips
- ✓ Dynamic Titles
- ✓ Parameter Table (Top N / Bottom N)
- ✓ Ranking Analysis
- ✓ Interactive Slicers
- ✓ Cross Filtering
- ✓ Custom Navigation Panel
- ✓ Power BI Service Publishing

Dashboard Pages

1 ☐ Executive Overview

Provides a high-level business summary.

KPIs

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Total Quantity
- Profit Margin %
- Return Rate %

Visuals

- Monthly Sales Trend
- Sales by Market
- Sales by Segment
- Sales by Country (Map)
- Profit by Category

2▣ Sales Performance

Provides detailed product and sales analysis.

Visuals

- Top N Products
- Sales vs Profit
- Customer Segment Analysis
- Decomposition Tree
- Dynamic Ranking

Features

- Top N Parameter
- Drill Down
- Dynamic Ranking

3▣ Customer Insights

Analyzes customer purchasing behavior.

Visuals

- Returned Orders Analysis
- Customer Distribution
- Market Analysis
- Customer Performance Table

Features

- Conditional Formatting
- Dynamic Filtering
- Customer Analysis

4▣ Time Intelligence

Provides historical sales comparisons.

Visuals

- Previous Year vs Current Year
- Rolling 12 Month Trend
- Sales by Year
- Decomposition Tree

Features

- YTD
- Previous Year
- Rolling 12 Months
- Time Intelligence

5 Drill-Through Page

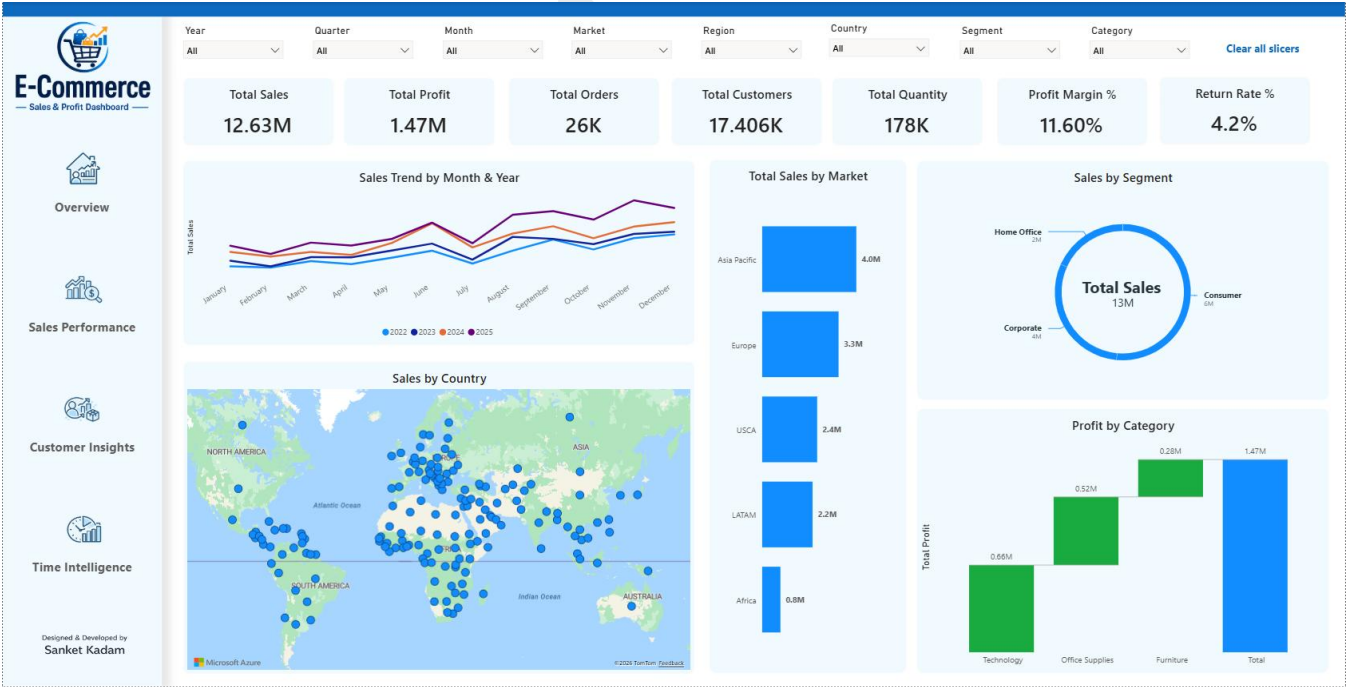
Dedicated drill-through page to analyze state-level performance.

Displays

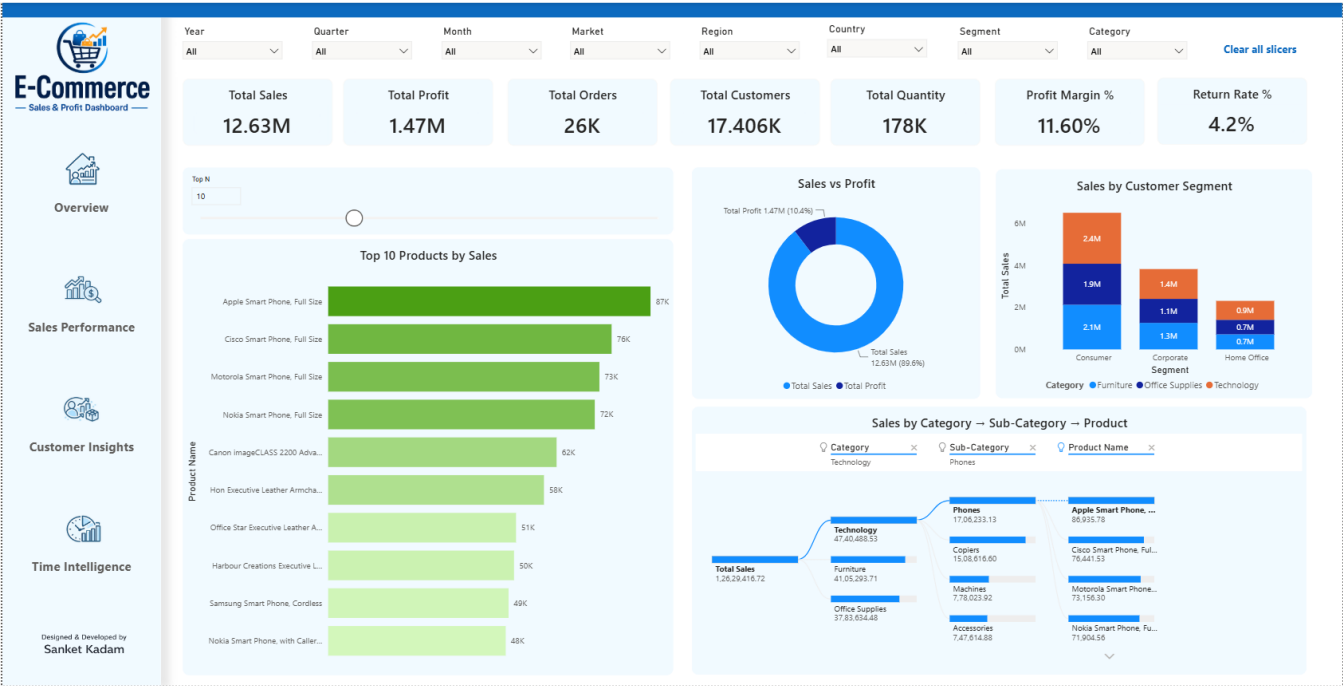
- Country
- Region
- State
- Sales
- Profit
- Orders
- Profit Margin

Dashboard Preview

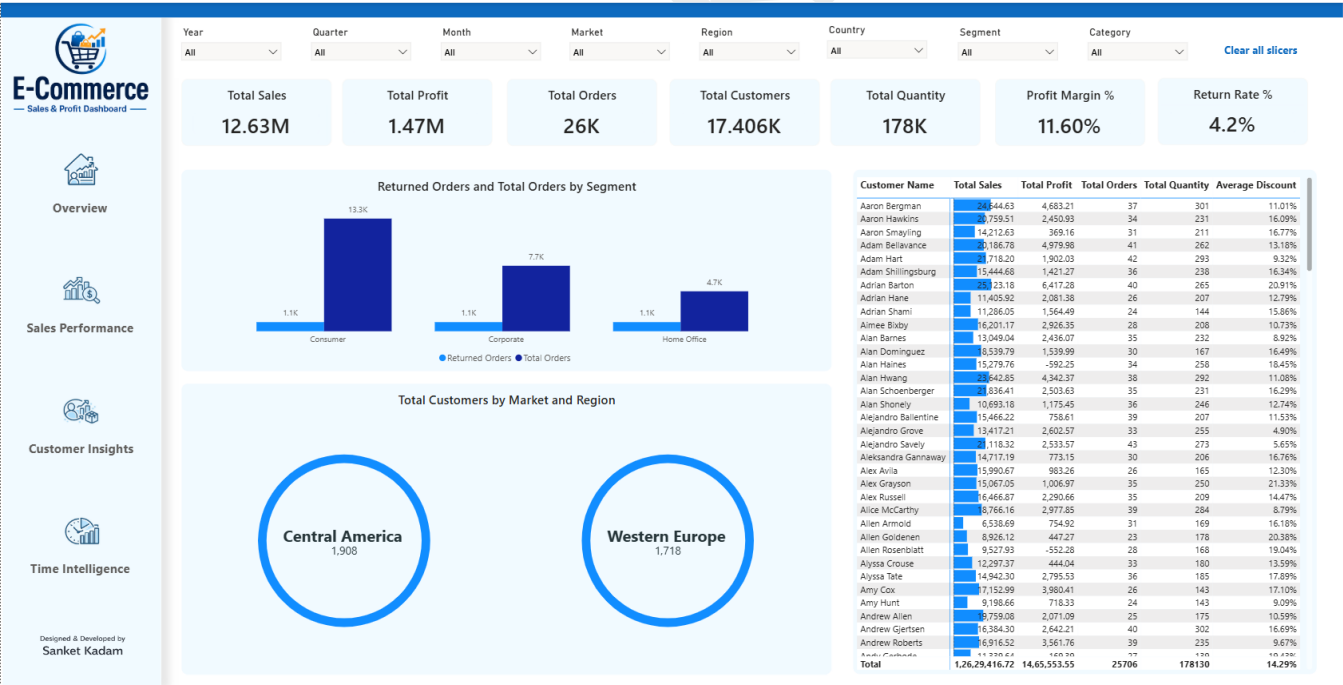
Executive Overview



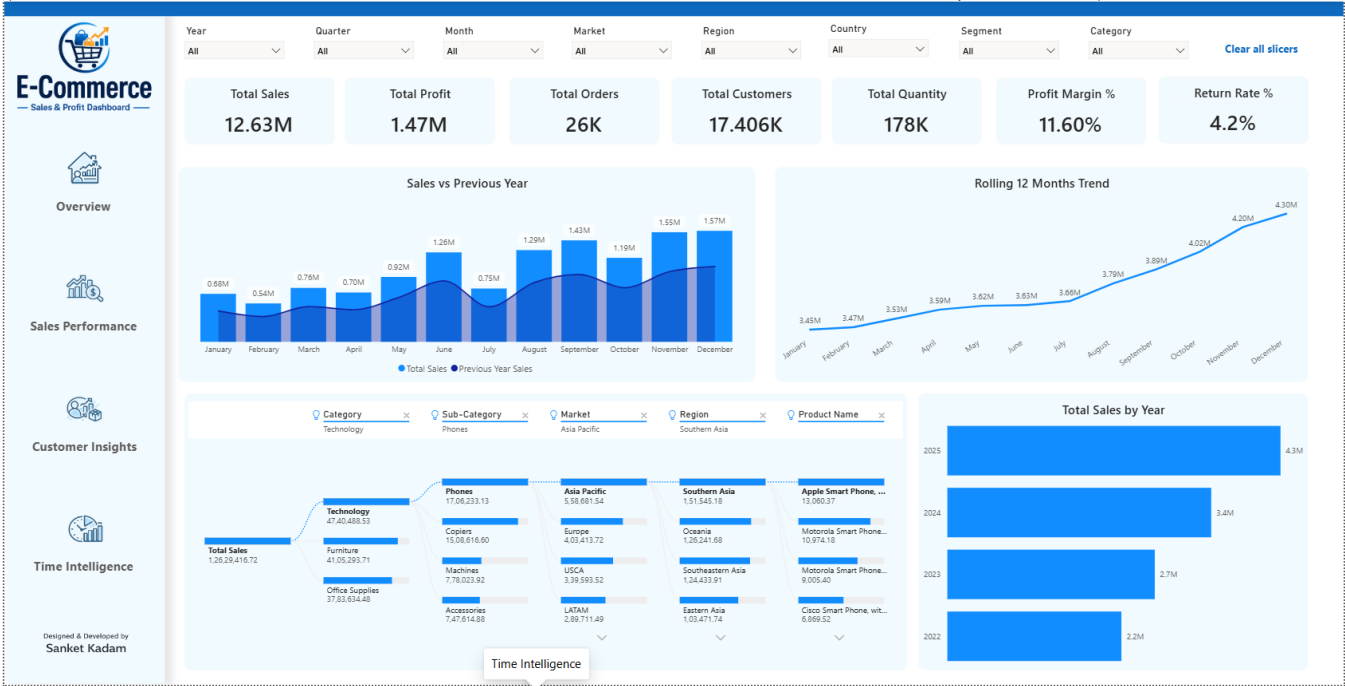
Sales Performance



Customer Insights



Time Intelligence



Drill Through

Country	Region	Market	State	Total Sales	Total Profit	Total Orders	Profit Margin %
Canada	Canada	USCA	Alberta	6,550.86	1,963.59	23	29.97%
Canada	Canada	USCA	British Columbia	9,544.38	2,695.47	24	28.24%
Canada	Canada	USCA	Manitoba	2,017.62	662.46	11	32.83%
Canada	Canada	USCA	Newfoundland	102.84	9.24	1	8.98%
Canada	Canada	USCA	Nova Scotia	382.32	40.80	3	10.67%
Canada	Canada	USCA	Ontario	34,104.96	8,767.83	103	25.71%
Canada	Canada	USCA	Quebec	10,924.17	2,961.42	31	27.11%
Canada	Canada	USCA	Saskatchewan	1,955.58	404.91	8	20.71%
Barbados	Caribbean	LATAM	Saint Michael	7,174.27	2,199.79	13	30.66%
Cuba	Caribbean	LATAM	Artemisa	2,370.20	836.74	7	35.30%
Cuba	Caribbean	LATAM	Camagüey	16,106.30	4,944.76	33	30.70%
Cuba	Caribbean	LATAM	Ciego de Ávila	6,385.38	1,799.62	16	28.18%
Cuba	Caribbean	LATAM	Cienfuegos	7,425.21	1,584.33	17	21.34%
Cuba	Caribbean	LATAM	Granma	17,002.54	3,970.82	31	23.35%
Cuba	Caribbean	LATAM	Guantánamo	10,942.86	2,131.02	25	19.47%
Cuba	Caribbean	LATAM	Holguín	13,223.39	2,797.85	32	21.16%
Cuba	Caribbean	LATAM	Isla de la Juventud	2,602.70	730.92	9	28.08%
Cuba	Caribbean	LATAM	Las Tunas	8,917.54	2,052.76	34	23.02%
Cuba	Caribbean	LATAM	Matanzas	4,722.19	1,003.19	19	21.24%
Cuba	Caribbean	LATAM	Mayabeque	7,073.87	1,633.75	8	23.10%
Cuba	Caribbean	LATAM	Pinar del Río	13,066.58	2,841.56	33	21.75%
Cuba	Caribbean	LATAM	Sancti Spiritus	3,436.03	909.37	10	26.47%
Cuba	Caribbean	LATAM	Santiago de Cuba	32,452.55	7,301.87	71	22.50%
Cuba	Caribbean	LATAM	Villa Clara	13,127.60	4,350.66	24	33.14%
Dominican Republic	Caribbean	LATAM	Barahona	4,118.37	513.51	5	12.47%
Dominican Republic	Caribbean	LATAM	Duarte	5,593.56	-707.58	16	-12.65%
Dominican Republic	Caribbean	LATAM	Españillat	1,039.12	-110.78	3	-10.66%
Dominican Republic	Caribbean	LATAM	La Romana	7,607.08	-61.20	26	-0.80%
Dominican Republic	Caribbean	LATAM	La Vega	2,686.41	-139.07	12	-5.18%
Dominican Republic	Caribbean	LATAM	Monseñor Nouel	159.10	-72.82	2	-45.77%
Total				1,26,29,416.72	14,65,553.55	25706	11.60%

Business Insights

This dashboard helps answer key business questions such as:

- Which products generate the highest sales?
- Which regions contribute the most profit?
- Which customer segment drives the most revenue?
- How have sales changed year-over-year?
- What is the return rate?
- Which products should be promoted?
- Which markets need improvement?

Skills Demonstrated

- Business Intelligence
- SQL Server
- Data Cleaning & ETL
- Data Modeling
- Star Schema
- SQL Views
- Power Query
- DAX
- Time Intelligence
- Interactive Dashboard Design
- Drill Through
- Drill Down
- Bookmarks
- Parameters
- Power BI Service
- Data Visualization

Developed By

Sanket Kadam

Senior Software Developer | Power BI Developer

Email: sanket.k1360@gmail.com

LinkedIn: <https://www.linkedin.com/in/sanketkadam1?>

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